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**Impact of molecular signal and population model
assumptions on the estimation of node heights in
time-resolved phylogenetic trees.**

Lab Rotation Report

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Abstract

Time-calibrated phylogenetic trees provide critical insights into pathogen evolution, transmission, and epidemiology, but their accuracy depends on both the availability of mutational signal and the choice of coalescent priors. In this study, we investigated how assumptions about population size models affect estimates of node heights in dated phylogenies. We derive analytic posteriors for the two-sequence tMRCA under constant size, exponential growth, exponential decline, and bottleneck models, and quantify prior influence using summary-based metrics such as mean shift, median shift, and mode shift, together with the Wasserstein-1 distance and the coalescent information ratio Ω . By mapping parameter regimes to exemplary pathogens, *Influenza* with high mutational signal and *Mycobacterium tuberculosis* with low signal, we show theoretically that prior impact declines with increasing mutational signal and effective population size. Yet, prior influence can be substantial when signal is weak or when population dynamics depart from constancy. Exponential growth can move the posterior to earlier or later times depending on the growth rate, whereas bottlenecks induce non-monotonic effects and secondary posterior peaks. We then simulate trees under constant and exponential growth population models, generate sequences, and re-estimate times on fixed topologies in BEAST using either a constant coalescent prior or a piecewise-constant skyline prior. The skyline prior recovers temporal population size changes and slightly reduces root-height and deep-branch errors under low signal with fast growth, whereas both priors perform similarly when signal is medium to high or when population models are more similar to constant models. Overall, demographic misspecification can bias time estimates in bacteria-like settings with limited signal. When the true demography is uncertain, flexible skyline priors are preferable, while constant priors remain adequate for high-signal viral analyses or large effective population sizes in near constant settings.

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Introduction

Reconstructing dated genealogies has become a powerful approach for uncovering the evolutionary history of pathogens and gaining key biological and medical insights. These time-calibrated phylogenetic trees reveal the timing and dynamics of pathogen emergence, spread, and adaptation. They have been instrumental in estimating the emergence of epidemiologically relevant lineages [1], reconstructing transmission histories between hosts [2], and inferring critical epidemiological parameters such as the basic reproductive number R_0 [3]. Moreover, dated genealogies enable the tracking of pathogen dispersal across geographic regions [4, 5, 6], the study of host-pathogen interactions, including the evolution of traits like antibiotic resistance [7, 8], and the identification of zoonotic spillover events [9]. As such, they are essential for reconstructing past outbreaks and guiding future public health strategies.

Reliable estimation of dated genealogies, however, requires sufficient genetic change over the sampling period. Populations, that accumulate sufficient mutations to provide a detectable temporal signal, are called measurably evolving populations (MEPs). Traditionally, this mainly included fast-evolving RNA viruses such as HIV and influenza, due to their high mutation rates and short generation times [10]. Through the increasing availability of whole genome sequencing, this concept has been extended to more slowly evolving pathogens, including bacteria and DNA viruses, where larger genome sizes help compensate for lower mutation rates [11, 12]. As a result, dated genealogies have become increasingly common for these pathogens [13], with *Mycobacterium tuberculosis* serving as a prominent example [6, 7, 14, 15]. Nevertheless, even when leveraging longer genomes, the mutational signal in such slowly evolving organisms remains weaker than in RNA viruses, making temporal inference more challenging.

The most popular approach to estimate dated phylogenetic trees from multiple sequence alignments and sampling times is through the application of the software BEAST [16]. BEAST implements a Bayesian framework that employs Markov Chain Monte Carlo (MCMC) sampling to jointly infer dated genealogies and phylodynamic parameters. It integrates the mutational signal through a likelihood function and incorporates prior information on model parameters to estimate their joint posterior distribution [17, 18].

Central to this inference is the coalescent prior, which describes how lineages trace

back to common ancestors. Kingman’s coalescent models this process as successive pairwise merging events, where effective population size determines the rate of coalescence: Smaller populations coalesce more quickly, while larger populations delay coalescence [19, 20].

The standard coalescent assumes a constant population size, but this is rarely realistic. Extensions of the model allow population size to vary over time [21, 22], enabling demographic histories such as exponential growth or bottlenecks to be incorporated. In practice, however, true population size trajectories are usually unknown. To address this, **BEAST** offers options to jointly estimate population sizes using models such as constant size, exponential growth, or logistic growth [17]. When the underlying demographic model is unknown, skyline methods approximate population size as a piecewise constant function, providing a flexible, data-driven alternative [23, 24]. Despite these options, constant coalescent priors remain popular because of their simplicity [15, 25], and widely used alternatives to **BEAST** such as **BactDating** [13] rely exclusively on them [26, 27, 28].

These practical constraints raise the question, how much wrongly assuming a constant coalescent prior affect the resulting dated genealogies, especially for pathogens with weak temporal signal. In this study, we address this question by combining theory and simulation. First, we derive analytical results to examine how constant, exponential growth, and bottleneck coalescent priors influence the posterior distribution of tMRCA estimates. We then use **BEAST** to infer dated phylogenies under both constant coalescent and skyline priors from datasets simulated under different demographic models, to validate these result.

Methods

Estimating posterior distribution of tMRCA between two samples

Instead of examining the impact of different tree priors on all branching times within a phylogenetic tree, we simplify the problem by focusing solely on the time to the most recent common ancestor (tMRCA) between two heterochronously sampled sequences. Under the infinite sites model, the genetic distance between these two sequences is represented by the number of mutations k separating them. Both sequences have length L and evolve under a strict substitution rate μ . The effective

population size is modeled as a function of time $N(t)$. Combining the coalescent prior on the tMRCA, $P(t_{\text{MRCA}}|N)$, with the likelihood of observing k mutations over time, $P(k|t_{\text{MRCA}})$, we obtain the posterior distribution on the t_{MRCA} :

$$P(t_{\text{MRCA}}|k) = \frac{P(k|t_{\text{MRCA}}) \cdot P(t_{\text{MRCA}}|N)}{P(k)} \propto P(k|t_{\text{MRCA}}) \cdot P(t_{\text{MRCA}}|N) \quad (1)$$

Likelihood representing mutational signal

The likelihood function follows a Poisson process with rate $\lambda = 2\mu L$, representing the number of mutations over time:

$$P(k|t_{\text{MRCA}}) = \frac{(2 \cdot \mu \cdot L \cdot t_{\text{MRCA}})^k}{k!} e^{-2\mu L t_{\text{MRCA}}} \quad (2)$$

The Maximum Likelihood Estimate of t_{MRCA} under this model is $\frac{k}{\lambda}$. Assuming a flat (uninformative) prior on t_{MRCA} , the posterior distribution of t_{MRCA} is proportional to the likelihood, yielding:

$$t_{\text{MRCA}} | k \sim \text{Gamma}(\text{shape} = k + 1, \text{rate} = 2\mu L) \quad (3)$$

The posterior mean and variance are then $\mathbb{E}[t_{\text{MRCA}}] = \frac{k+1}{2\mu L}$ and $\text{Var}[t_{\text{MRCA}}] = \frac{k+1}{(2\mu L)^2}$. Scaling both, the mutation count k and the rate λ by a factor $\alpha > 1$, reduces the variance and thus increases the mutational signal. Conversely, using a scaling factor $\alpha < 1$ decreases the mutational signal by increasing the variance.

Prior representing coalescent signal

While the flat prior is mathematically convenient, it ignores population dynamics that influence coalescent times. Under the coalescent, t_{MRCA} corresponds to the waiting time until the first coalescent event occurred. For a constant population size, the prior is exponentially distributed with rate $\frac{1}{N}$. This can be generalized to a time-varying population size:

$$P(t_{\text{MRCA}}|N) = \frac{1}{N(t_{\text{present}} - t_{\text{MRCA}}) \cdot g} \cdot e^{-\int_0^{t_{\text{MRCA}}} \frac{1}{g \cdot N(t_{\text{present}} - s)} ds} \quad (4)$$

We set the generation time $g = 1$ for simplicity and omit it in the following derivations.

Constant Population Size

The first demographic model we consider is a constant coalescent prior:

$$P(t_{\text{MRCA}} | N) = \frac{1}{N} e^{-\frac{1}{N} t_{\text{MRCA}}} \quad (5)$$

We investigate how this prior influences the posterior distribution of t_{MRCA} across different population sizes N , as well as under varying levels of mutational signal.

In this special case, the posterior is again Gamma-distributed:

$$t_{\text{MRCA}} | k, N \sim \text{Gamma} \left(\text{shape} = k + 1, \text{rate} = 2\mu L + \frac{1}{N} \right) \quad (6)$$

Exponential Growth

Next, we consider an exponentially growing population defined by $N(t) = N_0 \cdot e^{\beta t}$. The corresponding coalescent prior is:

$$P(t_{\text{MRCA}} | N) = \frac{1}{N_0 \cdot e^{\beta(t_{\text{present}} - t_{\text{MRCA}})}} \cdot e^{-\frac{1}{N_0 \cdot \beta} e^{-\beta t_{\text{present}}} (e^{\beta t_{\text{MRCA}}} - 1)} \quad (7)$$

Looking backward in time and setting $t_{\text{present}} = 0$, the population size declines as $N(s) = N_{\text{present}} \cdot e^{-\beta s}$. The prior simplifies to:

$$P(t_{\text{MRCA}} | N) = \frac{1}{N_{\text{present}} \cdot e^{-\beta t_{\text{MRCA}}}} \cdot e^{-\frac{1}{N_{\text{present}} \cdot \beta} (e^{\beta t_{\text{MRCA}}} - 1)} \quad (8)$$

Exponential Decline

Analogously, an exponentially declining population forward in time corresponds to exponential growth backward in time. In this case, the coalescent prior is:

$$P(t_{\text{MRCA}} | N) = \frac{1}{N_{\text{present}} \cdot e^{\beta t_{\text{MRCA}}}} \cdot e^{\frac{1}{N_{\text{present}} \cdot \beta} (e^{-\beta t_{\text{MRCA}}} - 1)} \quad (9)$$

Bottleneck

Lastly, we consider a population bottleneck, modeled with all times measured backward from the present. The parameters $\Delta t_{\text{b-start}}$ and $\Delta t_{\text{b-end}}$ denote the time intervals from the present to the onset and end of the bottleneck, respectively. The

population size is modeled as:

$$N(s) = \begin{cases} N_{\text{high}}, & \text{if } s < \Delta t_{\text{b-start}} \text{ or } s > \Delta t_{\text{b-end}}, \\ N_{\text{low}}, & \text{if } \Delta t_{\text{b-start}} < s < \Delta t_{\text{b-end}}. \end{cases} \quad (10)$$

The corresponding coalescent prior is defined as:

$$P(t_{\text{MRCA}}|N) = \begin{cases} \frac{1}{N_{\text{high}}} \cdot \exp\left(-\frac{t_{\text{MRCA}}}{N_{\text{high}}}\right) & \text{if } t_{\text{MRCA}} < \Delta t_{\text{b-start}} \\ \frac{1}{N_{\text{low}}} \cdot \exp\left(-\left(\frac{\Delta t_{\text{b-start}}}{N_{\text{high}}} + \frac{t_{\text{MRCA}} - \Delta t_{\text{b-start}}}{N_{\text{low}}}\right)\right) & \text{if } \Delta t_{\text{b-start}} < t_{\text{MRCA}} < \Delta t_{\text{b-end}} \\ \frac{1}{N_{\text{high}}} \cdot \exp\left(-\left(\frac{\Delta t_{\text{b-start}}}{N_{\text{high}}} + \frac{\Delta t_{\text{b-end}} - \Delta t_{\text{b-start}}}{N_{\text{low}}} + \frac{t_{\text{MRCA}} - \Delta t_{\text{b-end}}}{N_{\text{high}}}\right)\right) & \text{if } t_{\text{MRCA}} > \Delta t_{\text{b-end}} \end{cases} \quad (11)$$

Exemplary Pathogens Representing Low and High Mutational Signal

To contextualize our theoretical results with biologically relevant scenarios, we select parameter settings representative of two well-studied pathogens. As an example of a rapidly evolving RNA virus with a high mutational signal, we consider the hemagglutinin (HA) gene of *Influenza A*. In contrast, *Mycobacterium tuberculosis* serves as an example of a slowly mutating, largely non-recombinant bacterium with a comparatively large genome but overall lower mutational signal. To reflect situations where only partial genomes are available, we additionally consider scenarios for *M. tuberculosis* using half and quarter genome lengths, representing cases with an even weaker mutational signal. The selected parameters are summarized in Table 1 [14, 29, 30, 31, 32].

Table 1: Parameters for exemplary pathogen scenarios representing high and low mutational signals.

Pathogen scenario	Mutation rate μ	Genome length L	Expected t_{MRCA}
<i>Influenza</i> (HA gene)	3.5×10^{-3} /site/year	1,768 bp	2 years
<i>M. tuberculosis</i> (full genome)	4.5×10^{-8} /site/year	4,500,000 bp	200 years
<i>M. tuberculosis</i> (half genome)	4.5×10^{-8} /site/year	2,250,000 bp	200 years
<i>M. tuberculosis</i> (quarter genome)	4.5×10^{-8} /site/year	1,125,000 bp	200 years

Metrics to Evaluate Prior Impact on the Posterior Distribution

Median and Mode Shift

To quantify the influence of the prior on the posterior distribution, we compare the posterior obtained under a coalescent prior with the posterior resulting from a flat, uninformative prior (which is equivalent to the likelihood). This comparison is expressed in terms of the relative shift in the median or mode between the two distributions:

$$\delta_{\text{mode}} = \frac{\text{mode}_{\text{posterior}} - \text{mode}_{\text{likelihood}}}{\text{mode}_{\text{likelihood}}}, \quad \delta_{\text{median}} = \frac{\text{median}_{\text{posterior}} - \text{median}_{\text{likelihood}}}{\text{median}_{\text{likelihood}}}. \quad (12)$$

For general studies across different levels of mutational signal, these relative shifts provide a scale-free measure of prior influence. In contrast, when analyzing real pathogen data, the absolute time difference between medians or modes is often more directly interpretable.

In the special case of a constant coalescent prior, the shift in the posterior mean can be derived analytically. This yields:

$$\delta_{\text{mean}} = \frac{\mathbb{E}_{\text{posterior}}[t] - \mathbb{E}_{\text{likelihood}}[t]}{\mathbb{E}_{\text{likelihood}}[t]} = -\frac{1}{2\mu L N + 1}. \quad (13)$$

Relative Wasserstein Distance

The Earth Mover's or Wasserstein Distance is a metric used to quantify the dissimilarity between two probability distributions. Intuitively, it represents the minimum work required to transform one distribution into the other, where work is defined as the amount of probability mass that must be moved multiplied by the distance it is transported [33]. Formally, for two one-dimensional probability distributions $P(t)$ and $Q(t)$ with cumulative distribution functions (CDFs) F_P and F_Q , the first Wasserstein distance is given by:

$$W_1(P, Q) = \int_{-\infty}^{\infty} |F_P(t) - F_Q(t)| dt. \quad (14)$$

In our context, we compute the Wasserstein distance between the likelihood and posterior distributions using the `scipy.stats` implementation. To obtain a scale-free measure of prior influence, we normalize the resulting distance by the expected value of the likelihood.

Coalescent Information Ratio Ω

To quantify the relative contribution of the prior and the likelihood on the posterior, Parag et al. introduced the *Coalescent Information Ratio* Ω as an information-theoretic measure [34]. It is defined as:

$$\Omega = \sqrt{\frac{\det [\mathcal{I}(t_{\text{MRCA}})]}{\det [\mathcal{I}(t_{\text{MRCA}}) + \mathcal{P}(t_{\text{MRCA}})]}} \in [0, 1] \quad (15)$$

where $\mathcal{I}(t_{\text{MRCA}})$ is the Fisher information derived from the likelihood, dependent on the mutational data, and $\mathcal{P}(t_{\text{MRCA}})$ is the Fisher information contributed by the prior. A value of Ω close to 1 indicates that the posterior is primarily informed by the data and the prior has little influence, whereas a value close to 0 suggests that the prior dominates and the posterior is strongly shaped by prior assumptions. To compute Ω , we first evaluate the Fisher information of both the likelihood and the prior:

$$\mathcal{I}(t_{\text{MRCA}}) = \mathbb{E}_{k|t_{\text{MRCA}}} \left[\left(\frac{\partial}{\partial t_{\text{MRCA}}} \log P(k | t_{\text{MRCA}}) \right)^2 \right] = \frac{2\mu L}{t_{\text{MRCA}}}, \quad (16)$$

$$\mathcal{P}(t_{\text{MRCA}}) = \mathbb{E}_{t_{\text{MRCA}}|N} \left[\left(\frac{\partial}{\partial t_{\text{MRCA}}} \log P(t_{\text{MRCA}} | N) \right)^2 \right]. \quad (17)$$

Substituting different coalescent priors yields analytic expressions for Ω : Substituting different coalescent priors yields analytic expressions for Ω :

$$1. \text{ Constant: } \Omega(t_{\text{MRCA}}) = \sqrt{\frac{1}{1 + \frac{t_{\text{MRCA}}}{2\mu L N_0^2}}} \quad (18)$$

$$2. \text{ Exponential Growth/Decline: } \Omega(t_{\text{MRCA}}) = \sqrt{\frac{\frac{2\mu L}{t_{\text{MRCA}}}}{\frac{2\mu L}{t_{\text{MRCA}}} + \beta^2 + \frac{1}{N_0^2}}} \quad (19)$$

$$3. \text{ Bottleneck: } \Omega(t_{\text{MRCA}}) = \sqrt{\frac{\frac{2\mu L}{t_{\text{MRCA}}}}{\frac{2\mu L}{t_{\text{MRCA}}} + \mathcal{P}_{\text{bottleneck}}(t_{\text{MRCA}})}} \quad (20)$$

where the bottleneck prior information is piecewise constant with:

$$\mathcal{P}_{\text{bottleneck}}(t_{\text{MRCA}}) = \begin{cases} \frac{1}{N_{\text{high}}^2}, & t_{\text{MRCA}} < \Delta t_{\text{b-start}}, \\ \frac{1}{N_{\text{low}}^2}, & \Delta t_{\text{b-start}} < t_{\text{MRCA}} < \Delta t_{\text{b-end}}, \\ \frac{1}{N_{\text{high}}^2}, & t_{\text{MRCA}} > \Delta t_{\text{b-end}}. \end{cases} \quad (21)$$

Because all expressions for Ω depend on the time of evaluation, we compute Ω at the maximum a posteriori (MAP) estimate of t_{MRCA} .

Simulating Trees based on Population Size Models

For the simulation study, we generated 100 phylogenetic trees under three population size models: a uniform (constant) population size model, as well as slow and fast exponential growth models. For the uniform model, we assumed a constant population size of $N = 2000$. The exponential growth models were defined by trajectories of the form $N(t) = N(0) \cdot e^{-\beta t}$, where t is measured backward in time, $N(0) = 2000$ is the present-day population size, and β denotes the growth rate. We used $\beta = 0.001$ for slow exponential growth and $\beta = 0.002$ for fast exponential growth.

Tree simulations under these demographic models were performed using code adapted from [35], based on the `phylodyn` package [36].

Subsequently, sequence data were simulated along the generated trees using **Seq-Gen** [37] under a Jukes–Cantor mutation model with a mutation rate of $\mu = 4.6 \cdot 10^{-8}$, as reported for *M. tuberculosis*. To increase the mutational signal compared to the tuberculosis setting, we simulated sequences of length 10.000.000 bp.

Single nucleotide polymorphisms (SNPs) were extracted using **SNP-sites** [38], only retaining only mutated sites to reduce computational complexity. To generate datasets with varying levels of mutational signal, we considered three scenarios:

- High mutational signal: all SNPs included,
- Medium mutational signal: 5.000 SNPs, approximating the signal observed in *M. tuberculosis*,
- Low mutational signal: 1.000 SNPs.

In total, this resulted in 9 datasets, combining the three demographic models with the three levels of mutational signal.

Running BEAST on Simulated Trees

We then ran **BEAST** version 1.10.4 [39] on all simulated trees reestimating times on fixed phylogenies. A Jukes–Cantor substitution model was applied with a fixed molecular clock rate. The base clock rate of $\mu = 4.6 \cdot 10^{-8}$ was adjusted according to the number of SNPs included and the total simulated genome length as follows:

$$\mu' = \mu \cdot \frac{\text{number of included SNPs}}{\text{total number of generated SNPs}} \cdot \frac{10,000,000}{\text{number of included SNPs}}. \quad (22)$$

To evaluate whether wrongly assuming the population model to be constant introduces bias in time-calibrated tree estimation, we analyzed each simulated tree under a constant coalescent prior and a piecewise-constant skyline prior. For the constant coalescent prior, we specified an initial population size of 1.000 with a lognormal prior ($\mu = 0$, $\sigma = 4$). Chains were run for 10.000.000 steps. When using the skyline coalescent prior, ten population size groups were defined, each spanning an equal number of coalescent events. Each group used a lognormal prior with an initial population size of 1.000 ($\mu = 0$, $\sigma = 4$). Due to the increased complexity, chains were run for 80.000.000 steps.

In all cases, we sampled 10.000 states per run. Each analysis was repeated three times, and resulting log and tree files were combined using **LogCombiner** [40] with a 10% burn-in. Convergence was assessed by computing effective sample sizes (ESS) using the *checkESS* function from the **beastio** package [41]. Only runs with combined ESS values above 200 were retained for further analysis. Finally, we summarized posterior trees with **TreeAnnotator** [17], producing median height summary trees.

Evaluating difference in Population Size, Branch Length and Root Height

We first assess errors in population size by calculating the differences between estimated and true population sizes as well as their absolute and relative errors. The absolute, relative error is defined as

$$\Delta_{\text{abs, rel}} N = \frac{|N_{\text{est}} - N_{\text{true}}|}{N_{\text{true}}} \quad (23)$$

To evaluate the impact of misspecifying population models for coalescent priors, we compare estimates obtained under a constant coalescent prior with those from a skyline prior. Instead of node heights, we analyze branch lengths, since errors can accumulate across the tree when comparing node heights directly. As a corresponding time point, we choose the end point of each branch looking backwards in time. We compute branch length errors for both models while accounting for their timing and the associated population size errors.

As the root is the node most strongly affected by prior misspecification, we additionally compare root height estimates. Errors across trees within each condition are summarized using violin plots. To test whether root height errors differ significantly between the two models, we apply the non-parametric Wilcoxon signed-rank test (`scipy.stats`) with a significance level of 0.05.

Finally, we quantify the agreement between models by calculating the overlap of the 95% confidence intervals (CIs) of root height estimates under the constant coalescent and skyline priors. Since only CI bounds are available from the summary trees, we approximate the posterior distributions of root heights by normal distributions. The mean CI overlap is then computed across all trees for each condition.

Results

Dependence of Prior Impact on Mutational Signal

We first investigate the influence of the constant coalescent prior across population sizes $N = 10, 100, 1000, 10000$ and 100000 , while also varying the strength of the mutational signal through a scaling factor α . As shown in Figure 1, larger population sizes and stronger mutational signals both reduce the influence of the prior. This trend is consistent across all evaluated metrics, including the relative Wasserstein-1 distance, the relative median shift, and the reverse coalescent information ratio $1 - \Omega$.

We then extend this analysis to all population models and relate the findings to biologically realistic scenarios (Figure 2). Parameter choices were informed by two well-studied pathogens: *Influenza*, representing a rapidly evolving RNA virus with high mutational signal, and *Mycobacterium tuberculosis*, representing a slowly evolving bacterium with a low mutational signal. To further explore weaker mutational signals, we additionally consider scenarios using half and quarter genome lengths of *M. tuberculosis*.

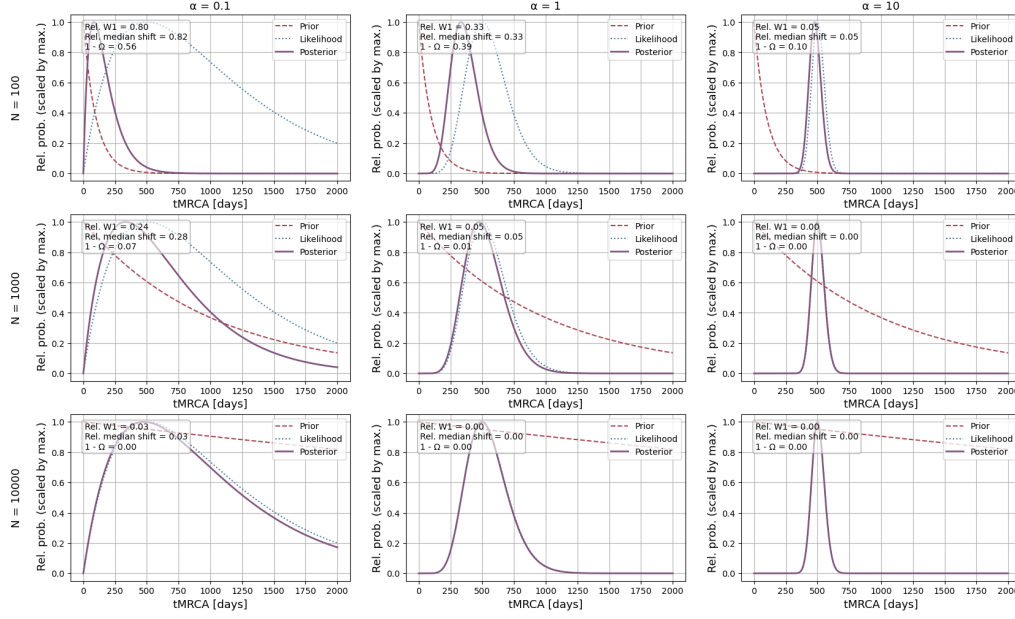


Figure 1: Impact of Constant Coalescent Prior with varying population size N and mutational signal. Unnormalized likelihood, prior, and posterior distributions of t_{MRCA} were computed and rescaled to the interval $[0, 1]$ by their respective maxima. Scenarios compare different population sizes N and scaling factors α that modulate the mutational signal. Larger population sizes and stronger mutational signals reduce the influence of the prior. Prior impact was quantified using the relative Wasserstein distance, relative median shift, and the reverse coalescent information ratio $1 - \Omega$.

Across all population models, the prior has negligible influence for *Influenza*, whereas for *M. tuberculosis*, its impact becomes increasingly pronounced as the mutational signal decreases. Figure 2 further illustrates how prior influence depends on key model parameters, such as the transmission rate in exponential models and bottleneck duration in bottleneck models. Importantly, these trends remain consistent across all pathogen scenarios.

These findings underscore how both population demography and mutational signal jointly determine the degree to which the prior shapes posterior estimates of t_{MRCA} . The following section examines the influence of these parameter choices in greater detail.

Impact of Parameter Choices for Different Population Models

Since prior influence was negligible for *Influenza*, we focus our analysis on the *M. tuberculosis* scenario, where the lower mutational signal allows for clearer observation of prior effects. Figure 3 summarizes the influence of coalescent priors across the four demographic models using the median shift as a representative metric of

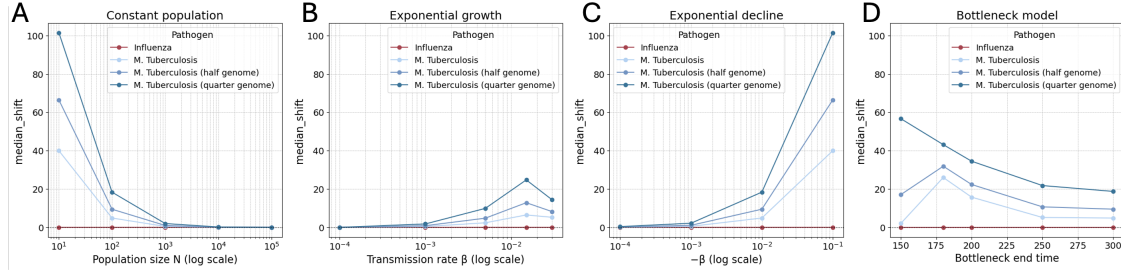


Figure 2: Impact of Mutational Signal on Prior Contribution. Comparison of four scenarios representing decreasing mutational signal: *Influenza*, *M. tuberculosis* (full genome), and *M. tuberculosis* with half and quarter genome lengths. Prior impact is quantified by the median shift between the likelihood and posterior distributions. For the high mutational signal case (*Influenza*), the prior has negligible influence, whereas for lower mutational signal cases (*M. tuberculosis*), the prior exerts a stronger effect. (A) Under a constant population size model, prior impact diminishes with increasing population size N . (B) In the exponential growth model (with $N = 10,000$), prior impact depends on the growth rate β . (C) In the exponential decline model (also with $N = 10,000$), faster decline (more negative β) increases prior impact. (D) For the bottleneck model ($N_{\text{high}} = 1000$, $N_{\text{low}} = 100$), prior impact is dependent on the bottleneck duration.

prior impact.

Under a constant coalescent prior (Figure 3A), increasing either the population size N or the mutational signal reduces prior influence. Once N exceeds approximately 1000, the median shift becomes negligible, and the posterior is almost entirely driven by the likelihood. This aligns with earlier findings (Section 1), where stronger mutational signals similarly reduced prior dependence.

For an exponentially growing population model (Figure 3B), prior influence is more nuanced. As shown in Figure 4, exponential growth can shift the posterior in either direction depending on the growth rate β . Fast growth rates move the prior peak before the likelihood peak, pulling the posterior closer to the present. In contrast, moderate growth emphasizes historically smaller population sizes, shifting the posterior further back in time. For very small β , the prior approaches the constant coalescent prior, minimizing its impact.

The exponentially declining population prior (Figure 3C) consistently shifts the posterior towards more recent times, reflecting the increased probability of coalescent events under shrinking population sizes. This effect becomes more pronounced for larger negative values of β , as shown in Supplementary Figure 9.

For the bottleneck model (Figure 3D), we observe a non-monotonic effect. Short bottlenecks exert only minor influence, producing a secondary peak in the pos-

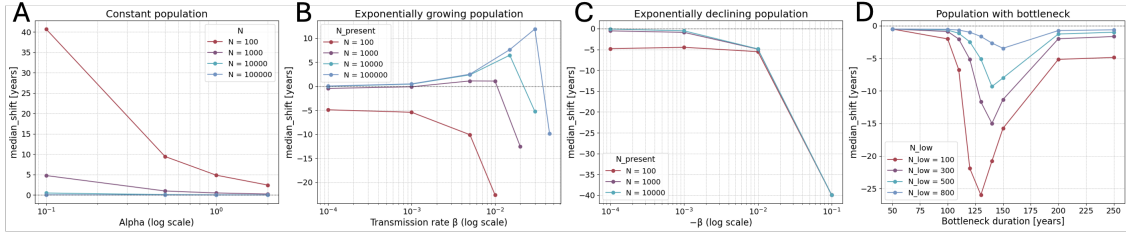


Figure 3: Summary of prior impact across population models for the *M. tuberculosis* scenario. The influence of different coalescent priors on the posterior distribution of t_{MRCA} is quantified using the median shift. (A) Under a constant coalescent prior, higher mutational signal and larger population sizes N reduce the median shift. For $N > 1000$, prior influence becomes negligible. (B) For an exponentially growing prior, the posterior can shift in either direction depending on the growth rate β . (C) An exponentially declining prior consistently shifts the posterior towards more recent times, reflecting reduced effective population size in the past. (D) For a bottleneck prior, prior influence peaks at intermediate bottleneck durations. Here, the bottleneck begins 50 years ago, with a population size outside the bottleneck of $N_{\text{high}} = 1000$.

terior without substantially shifting its median or mode. As bottleneck duration increases, its impact intensifies, and the secondary peak can overtake the likelihood peak, strongly shaping the posterior. For very long bottlenecks, however, the prior behaves similarly to a constant prior with reduced population size, and its influence decreases again.

The median shift shown in Figure 3 is used as a representative metric. Nevertheless, prior influence was systematically assessed using multiple complementary metrics, as illustrated in Supplementary Figure 10 for the exponential growth case. Summary-based metrics (mean, median, mode shift) agree closely and have the advantage of indicating the direction of the shift, with negative values denoting movement toward more recent times. In contrast, distributional metrics such as the relative Wasserstein-1 distance and the reverse coalescent information ratio ($1 - \Omega$) provide absolute measures of prior influence without directional information but are more sensitive to subtle differences in distributional shape. In this study, all metrics exhibited strong agreement, which justifies the use of the median shift as a representative measure in the main analysis.

Together, these findings highlight that prior influence on posterior estimates of t_{MRCA} depends strongly on both the population model and its parameterization. While constant and slow-growth priors exert predictable effects that diminish with higher mutational signal and larger N , more complex models such as exponential growth or bottlenecks can shift the posterior in nuanced and sometimes opposing ways,

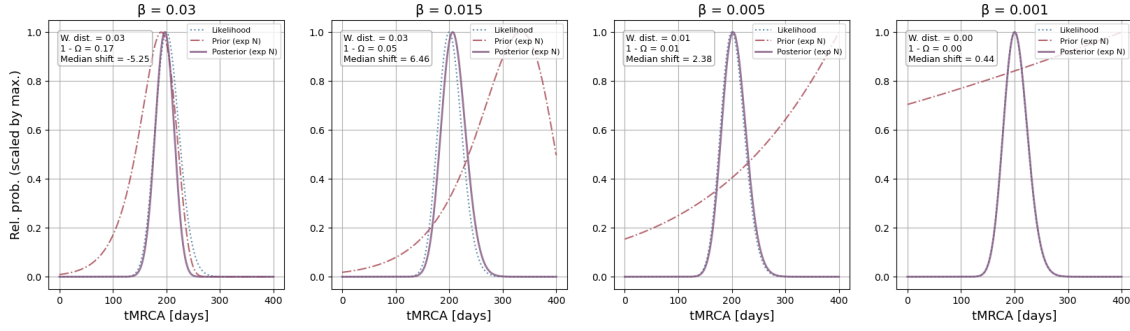


Figure 4: Impact of Exponentially Growing Coalescent Prior with varying rate β on t_{MRCA} posterior. We consider a present-day population size of $N_{\text{present}} = 10000$ and the *M. tuberculosis* scenario. The coalescent prior and posterior under exponential growth are shown in red and purple, respectively, while the likelihood function is shown in blue. Exponential growth can shift the posterior in both directions, depending on the rate β . Rapid growth places the prior peak earlier than the likelihood peak, pulling the posterior closer to the present. Moderate growth, in contrast, emphasizes smaller historical population sizes, shifting the posterior further back in time. For very slow growth, the prior approaches the constant coalescent prior, minimizing its influence on the posterior.

emphasizing the importance of carefully selecting demographic priors in practical analyses.

Evaluating Population Size Errors over Time

After establishing theoretical expectations for the impact of different population models and parameters, we evaluated these findings in practice by conducting a simulation study. We simulated trees under three population models: uniform, slow exponential growth, and fast exponential growth, each at low, medium, and high mutational signal. To assess the effect of misspecifying the coalescent prior, we compared results from BEAST runs using either a constant coalescent prior or a skyline prior.

As a first result, we examine the estimated population sizes (Figure 5). For trees simulated under a uniform population size, both the skyline and constant coalescent priors reproduce the true population size well on average, although the skyline model shows higher variance. In contrast, for the exponential growth scenarios, the constant coalescent prior systematically overestimates population sizes, while the skyline prior captures the decline in population size backward in time. The skyline model aligns most closely with the true curve during the last 500 years but tends to overestimate population sizes at earlier time points. This bias arises because fewer branching events occur further back in time, forcing the model to average over longer

intervals. Posterior samples further reveal that variance is highest at the most recent time points and decreases toward the past, reflecting the limited mutational signal in short, recent branches.

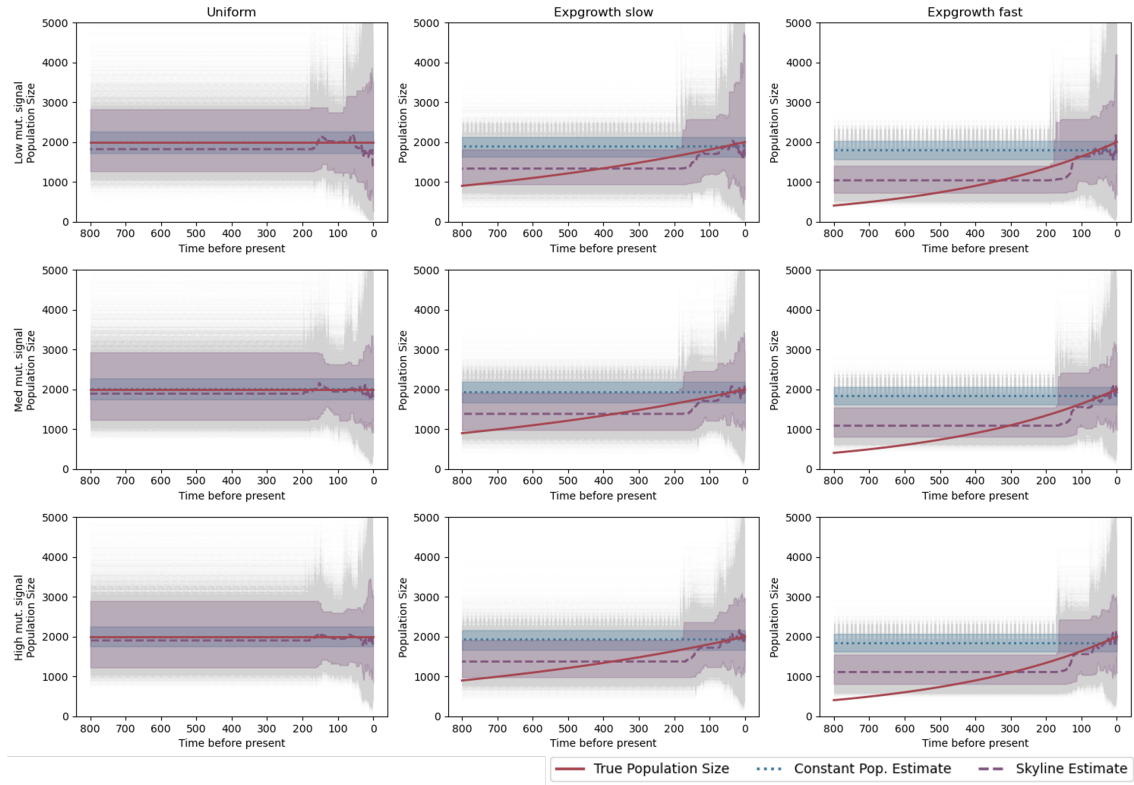


Figure 5: True and estimated population size for skyline and constant coalescent priors over time. Results are shown for all simulation settings (uniform, slow exponential growth, and fast exponential growth population models; low, medium, and high mutational signal). Estimates obtained under the skyline prior are shown in purple, with the dashed line indicating the median across 100 simulated trees and the shaded area the corresponding 95% confidence interval. Analogous results for the constant coalescent prior are shown in blue. Grey lines display population size trajectories from 100 posterior samples for each of the 100 simulated trees, visualizing the variance under both priors. The true population size is shown as a red line.

Comparing Errors in Root Height between Constant Coalescent and Skyline Priors

We next compare differences in root height between estimated and true trees under the skyline and constant coalescent priors. Figure 6 summarizes the relative absolute errors across the 100 trees simulated for each setting.

As expected, for the uniform population model, no systematic differences between

the two priors can be observed. Errors are largest in this setting, likely because the uniform model produces root times further in the past compared to the exponential growth models. For the exponential growth scenarios, no significant differences emerge under medium and high mutational signal. However, under low mutational signal, the skyline prior yields slightly lower errors, with the effect being most pronounced in the fast exponential growth setting where we expect prior misspecification to have the strongest impact. To further quantify the patterns observed in

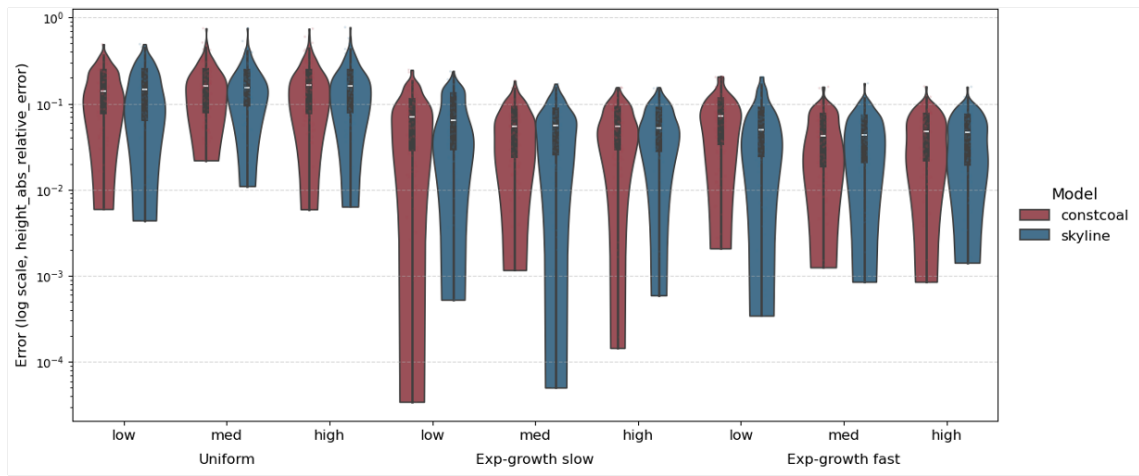


Figure 6: Relative absolute error in root height for constant coalescent and skyline priors. Violin plots show the distribution of relative absolute errors between estimated and true root heights (log scale) across 100 trees per setting. Results for constant coalescent priors are shown in red and for skyline priors in blue.

Figure 6, we performed Wilcoxon signed-rank tests and calculated the overlap of the 95% confidence intervals (CIs) of root height estimates across all settings. The Wilcoxon tests indicated significant differences for the high mutational signal under both slow and fast exponential growth, as well as for the low mutational signal under fast exponential growth.

Table 2 reports the mean CI overlap values. For high mutational signal, overlaps remain close to one across all population models, indicating almost no difference between skyline and constant coalescent priors. Even though all overlap values are above 80%, the overlap consistently decreases with decreasing mutational signal, reflecting greater divergence between the two priors. The lowest overlap is observed under low mutational signal with fast exponential growth, followed by the corresponding low signal case for slow exponential growth.

Table 2: Mean overlap of 95% confidence intervals of root height estimates between skyline and constant coalescent priors.

	Uniform			Slow Exp. Growth			Fast Exp. Growth		
Mutational Signal	Low	Med	High	Low	Med	High	Low	Med	High
CI Overlap	0.914	0.956	0.990	0.895	0.960	0.997	0.892	0.965	0.995

Evaluating Errors in Branch Length

To gain a more detailed perspective, we analyze errors in branch length across all branches in a tree rather than focusing only on the root. For visualization, we bin branch lengths from all trees of each setting by the time of their endpoints. Figure 7 compares branch length errors between the skyline and constant coalescent priors in relation to the corresponding errors in population size. For the exponential growth models, larger deviations in population size coincide with higher branch length errors for both priors. There is also a slight tendency toward higher errors under the constant coalescent prior in the low mutational signal case, and errors generally increase for deeper time points.

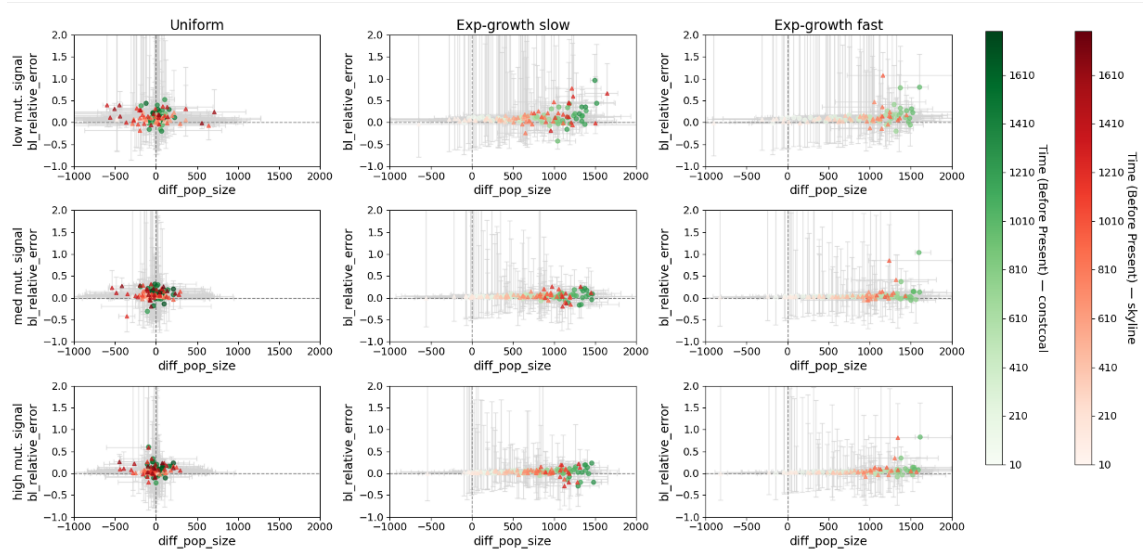


Figure 7: Branch length error vs. population size deviation over time. Errors in branch length across all branches and trees are binned into time intervals of 20 years. For each bin, median relative branch length errors are plotted against the median deviation in population size. Results obtained using a skyline prior are shown as red triangles, and those from a constant coalescent prior as green circles. Error bars indicate the 95% confidence intervals of both branch length and population size error within each bin.

Supplementary Figure 11 further illustrates these effects by showing violin plots of broader time bins along with the corresponding population size errors. Recent

times display greater variance but a lower median branch length error, whereas earlier times exhibit a higher median error with reduced variance. In the fast exponential growth setting with low mutational signal, the skyline prior yields lower errors than the constant coalescent prior at later time points. However, no consistent relationship is observed between the time window where the skyline population size estimate best matches the truth and the corresponding branch length error.

Discussion

Our theoretical analysis demonstrates that choosing appropriate population models can affect the estimation of dated phylogenetic trees. However, the impact is strongly dependent on the amount of mutational signal available and the underlying population dynamics. For settings with high mutational signal, such as rapidly evolving viruses like *Influenza*, we found essentially no differences between coalescent priors. This suggests that assuming a constant population size is not problematic in viral phylodynamics. A similar conclusion holds for bacteria under constant population size when the effective population size exceeds 1000, even if its exact value is unknown.

In contrast, for bacteria with more complex demographic histories, prior misspecification clearly had an effect. This was evident in the case of *Mycobacterium tuberculosis*, where the choice of prior influenced the posterior distribution of the tMRCA under models with small effective population size, exponential growth or decline, or bottlenecks. This effect was amplified when only subsets of the genome were analyzed, further reducing the mutational signal. In addition, incorrect assumptions about model parameters, such as growth rate or bottleneck duration, biased inference. Interestingly, exponential growth priors were able to shift estimates either forwards or backwards in time depending on the transmission rate. Bottleneck models produced even more complex behaviors. They generated secondary peaks that could remain minor, become dominant, or merge into a single peak, effectively mimicking a constant population model with smaller N , depending on their duration. These findings highlight that while constant or slow-growth priors tend to exert predictable effects that diminish with stronger signal and larger N , more complex priors can alter posterior estimates in more nuanced and sometimes opposing ways.

All theoretical results were obtained only considering the tMRCA of two homochronously sampled sequences. Extending the analysis to heterochronous sampling or complete trees would provide further insight. Whole-tree analyses would also

make it possible to investigate richer demographic models such as periodic population dynamics, as observed in influenza due to its seasonality [42], which are averaged out when considering only pairs of sequences.

In practice, the true demographic model is often unknown. Skyline priors therefore provide a useful alternative, as they enable co-estimation of population size trajectories [43, 24, 34]. In our simulation study, we tested whether skyline priors improve inference when the true population size is not constant. As expected, only skyline priors were able to approximate the true population size well for exponentially growing populations. We also observed a tendency toward improved reconstruction of phylogenetic times under exponential growth, with differences most pronounced in the challenging setting of low mutational signal and fast growth. In this case, skyline priors reduced root height errors and showed a trend toward improved branch length estimates, particularly for deeper branches. This represents an extreme case of limited signal, with even less information than in the tuberculosis example. Such scenarios may arise in practice when analyses are restricted to single genes or partial bacterial genomes, for example when excluding recombinant regions in estimating dated phylogenies as proposed in [44, 45].

Nevertheless, for slow exponential growth and settings with higher mutational signal, skyline and constant priors performed similarly. A more finely binned skyline prior might better recover the true population dynamics, potentially revealing larger improvements in root height or branch length estimation in these cases. Testing scenarios with faster growth rates, or heterochronous sampling could further increase the observable impact of prior misspecification.

Future work could also explore additional demographic scenarios, such as bottlenecks (as observed for *Bordetella pertussis* during the COVID-19 pandemic [46]) or seasonal fluctuations (as in influenza [42]). Moreover, applying alternative coalescent priors available in BEAST, such as explicit exponential growth models, could enhance phylodynamic reconstruction and provide further insight into the effects of prior misspecification.

Overall, our results demonstrate that demographic misspecification can substantially affect phylogenetic inference in theory, particularly in bacterial systems with limited signal or complex population histories. Our simulation study revealed similar tendencies, but further work is needed to determine whether assuming a constant population size introduces significant biases in practical bacterial analyses. As a precaution, we recommend avoiding overly restrictive coalescent models and instead

using skyline priors, which provide greater flexibility and reduce the risk of systematic errors in bacterial settings.

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Supplement

Effects of Coalescent Prior

Bottleneck

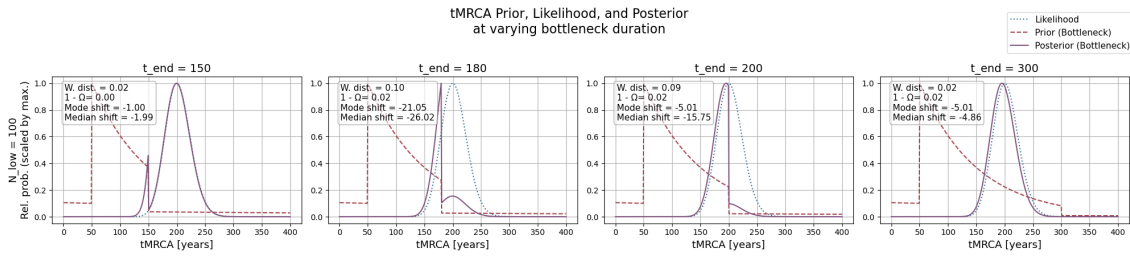


Figure 8: Impact of Bottleneck Coalescent Prior with varying bottleneck duration on t_{MRCA} posterior. We consider the *M. tuberculosis* scenario, with a population size of $N_{\text{high}} = 1000$, a bottleneck population size of $N_{\text{low}} = 100$, and a bottleneck onset 50 years in the past. Short bottlenecks introduce a secondary peak in the posterior but have little effect on its median or mode. As the bottleneck duration increases, this secondary peak becomes dominant, substantially shifting the posterior. For even longer bottlenecks, the prior approximates a constant coalescent prior with a reduced population size, which decreases its influence on the posterior once again.

Exponential Decline

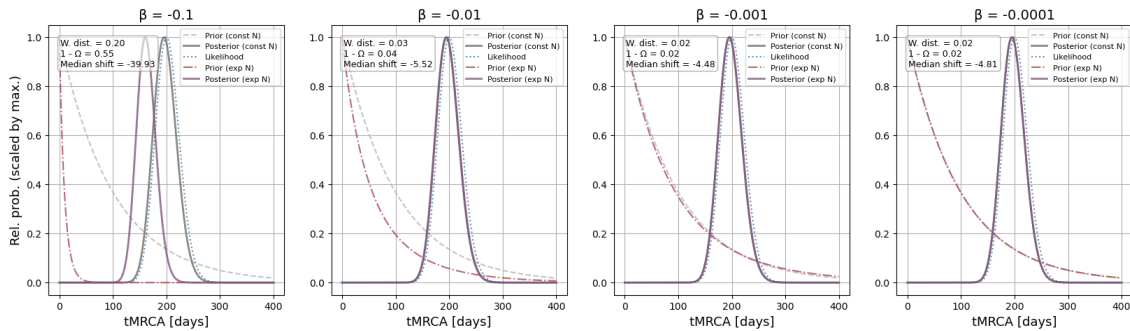


Figure 9: Impact of Exponentially Declining Coalescent Prior with varying rate β on t_{MRCA} posterior. We consider a present-day population size of $N_{\text{present}} = 100$ and the *M. tuberculosis* scenario. The coalescent prior and posterior under exponential decline are shown in red and purple, respectively, while the constant coalescent prior and posterior are shown in gray for comparison. The likelihood function is plotted in blue. Because exponential decline corresponds to exponential growth when viewed backward in time, it reduces the probability of late coalescent events. As the rate of decline β becomes more negative, the posterior shifts closer to the present, reflecting stronger prior influence.

Exemplary metrics comparison for exponential growth

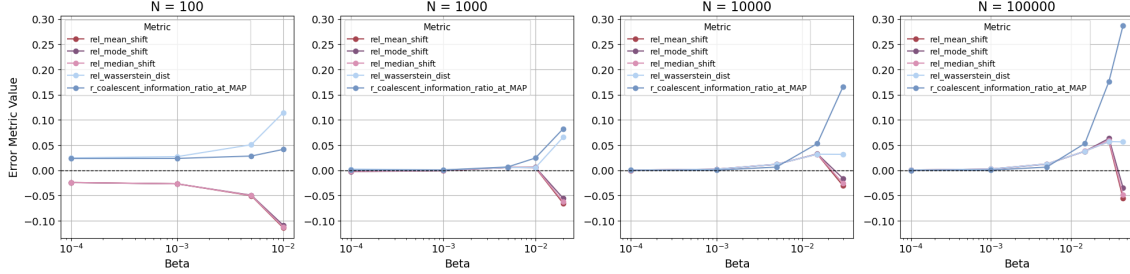


Figure 10: Comparison of prior impact metrics for an exponential growth coalescent prior. We compare several metrics used to quantify the influence of the prior on the posterior distribution of t_{MRCA} : relative mean shift, relative median shift, relative mode shift, relative Wasserstein-1 distance, and the reverse coalescent information ratio ($1 - \Omega$). Metrics based on summary statistics (mean, median, mode) tend to be consistent with each other and have the advantage of indicating the direction of the shift. Negative values indicate a shift towards more recent times. In contrast, the Wasserstein distance and reverse coalescent information ratio provide absolute measures of prior influence without directional information but capture distributional differences more comprehensively. Results are shown for varying growth rates β across different population sizes N . While all metrics agree qualitatively on the trend that faster growth increases prior influence, absolute measures such as the Wasserstein distance and $1 - \Omega$ exhibit stronger sensitivity to changes in β , particularly for large population sizes.

Error in Branch Length and Population Size over Time

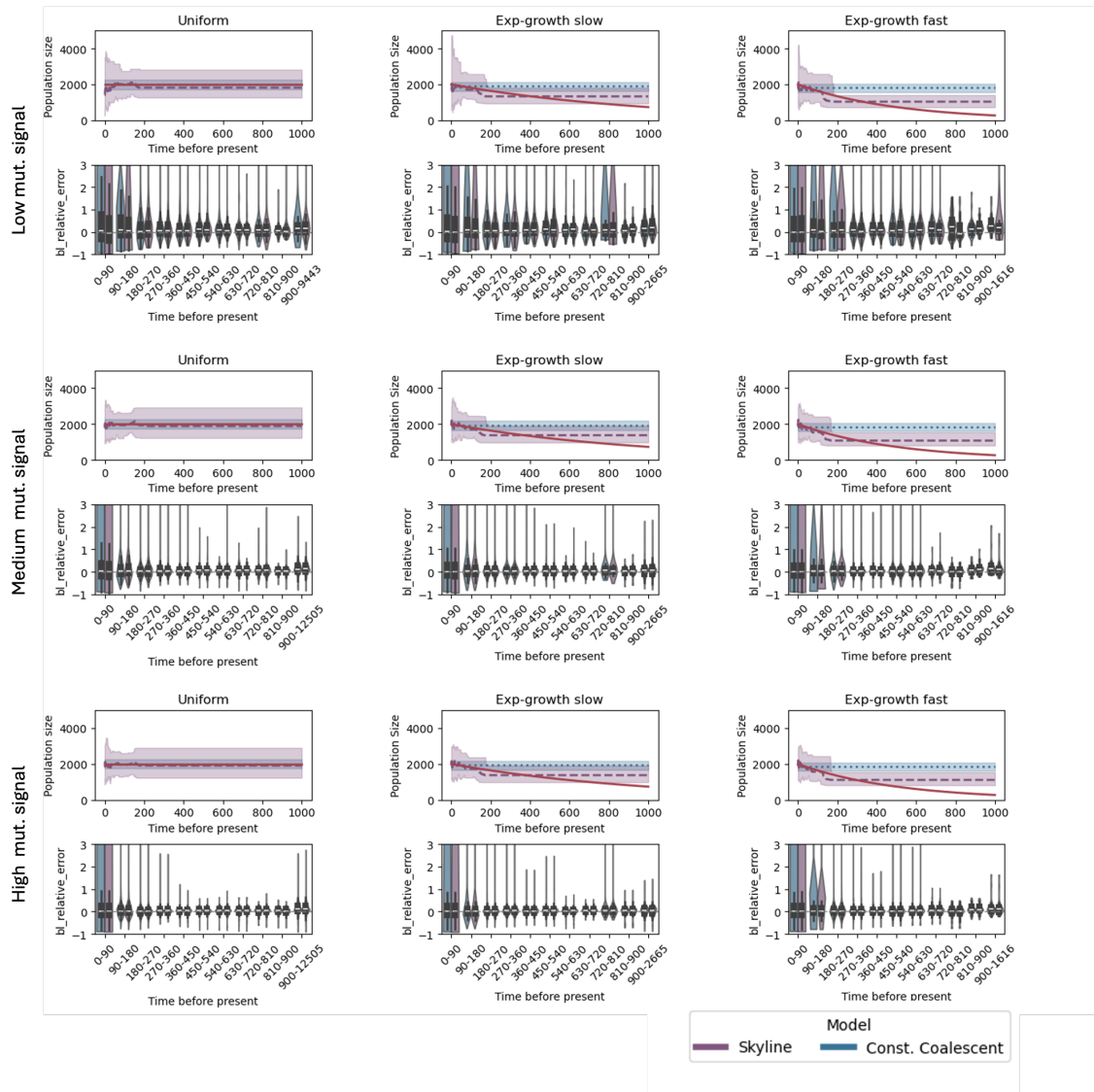


Figure 11: Branch length and population size errors over time. Errors are shown on a common time axis for all settings, skyline prior in purple and constant coalescent prior in blue. Recent times exhibit high variance but a low median branch length error, whereas deeper times show a higher median error with lower variance. Under fast exponential growth with low mutation signal, the skyline prior yields lower errors than the constant coalescent prior for late time points. We do not observe a consistent relationship between the time window where the estimated skyline population size matches the truth best and the branch length error.